B.Tech. CSE VI CSS 336

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING LAB

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Embedded Systems Laboratory

Assignment- II

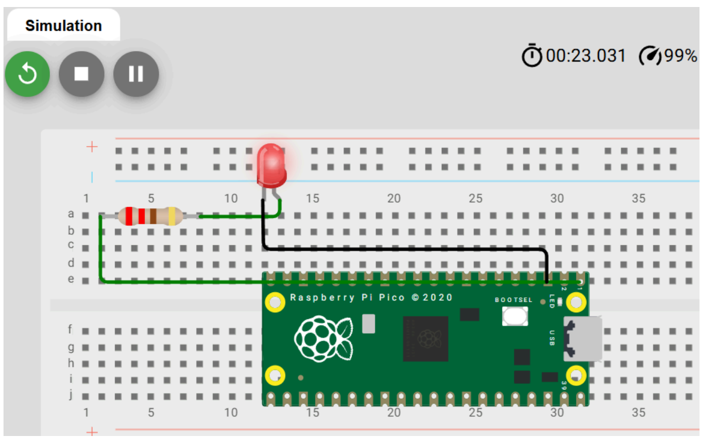
January 28, 2026

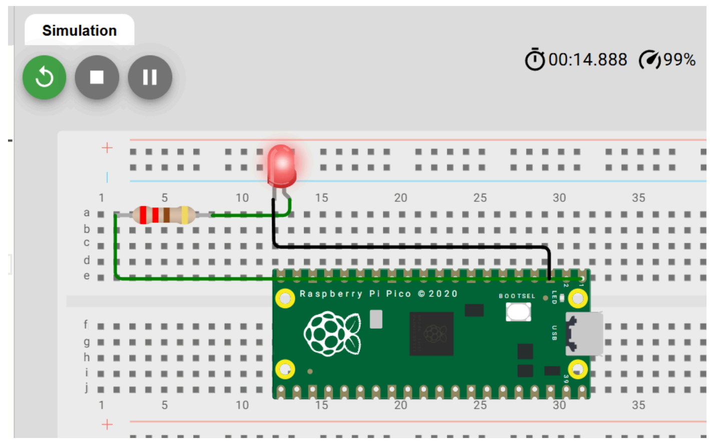
1. **Simulate the sending of a SOS message using Morse Code. The message is coded as 3D-3d-3D (where D=Dot and d=Dash), A dot is flashed by keeping the LED on for 0.25s and a dash is signalled by keeping the LED on for 1 sec. The delay between individual signals is 0.2 seconds. You can assume the signal is sent continuously**

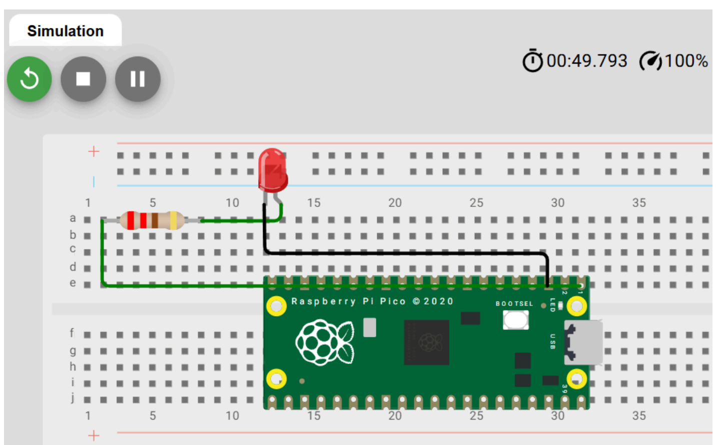
   Code

```
from machine import Pin
import time

led = Pin(0, Pin.OUT)

def dot():
    led.on()
    time.sleep(0.25)
    led.off()
    time.sleep(0.2)

def dash():
    led.on()
    time.sleep(1)
    led.off()
    time.sleep(0.2)

while True:
    # S : D D D
    dot(); dot(); dot()
    time.sleep(0.5)
```

```
# O : d d d
dash(); dash(); dash()
time.sleep(0.5)

# S : D D D
dot(); dot(); dot()
time.sleep(1)
```





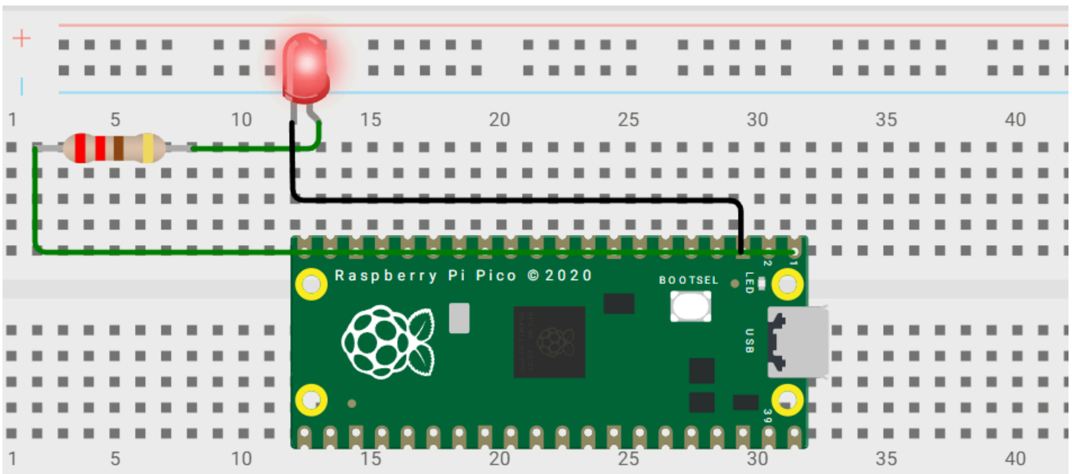


2. A cafe owner has a cafe named "7". He want the name of the cafe to blink on once every 4 seconds, stay on for a second and then blink off. Simulate the same using your pico.

Code

```
from machine import Pin
import time
```

```
led = Pin(0, Pin.OUT)

while True:
    led.on()
    time.sleep(1)
    led.off()
    time.sleep(3)
```

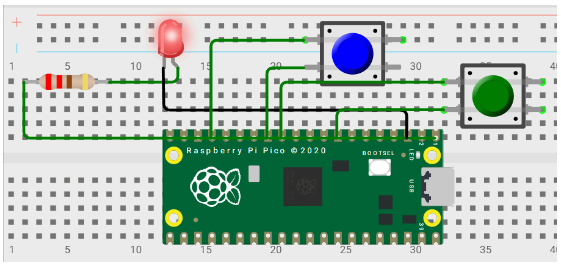


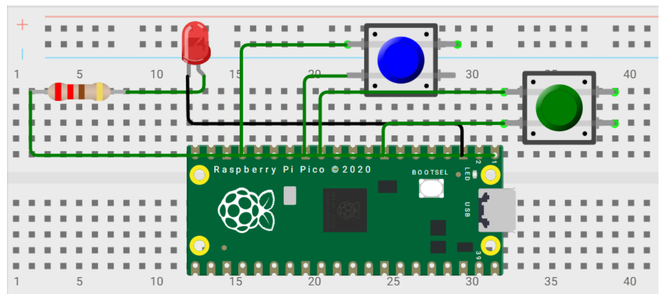
3. Simulate a system having a LED and two buttons. Pressing the first button speeds up the blinking of the LED, while pressing the second button slows down the blinking of the LED.

Code

```
from machine import Pin
import time

led = Pin(0, Pin.OUT)
btn_fast = Pin(1, Pin.IN, Pin.PULL_UP)
btn_slow = Pin(2, Pin.IN, Pin.PULL_UP)

delay = 0.5

while True:
    if not btn_fast.value():
        delay = max(0.1, delay - 0.1)
        time.sleep(0.1)
```

```
    if not btn_slow.value():
        delay += 0.1
        time.sleep(0.6)

    led.toggle()
    time.sleep(delay)
```

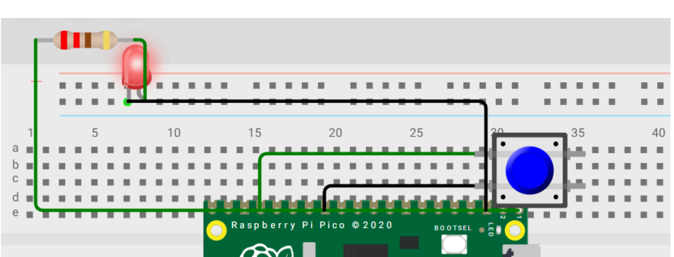


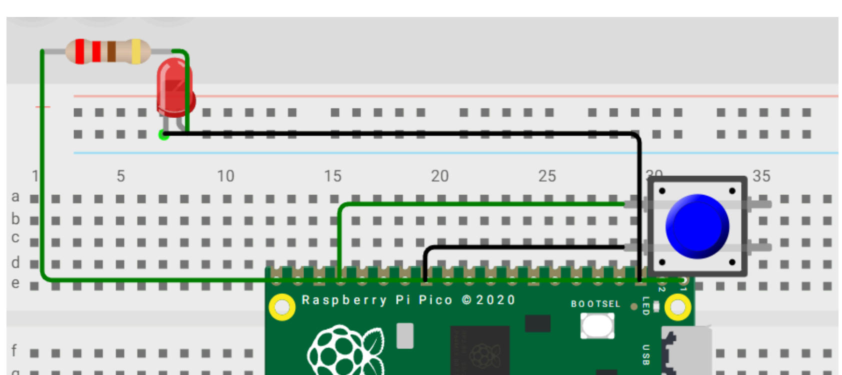


4. Design a switch LED system to turn on a LED when a switch is pressed and turn it off on the next press of the button.

Code

```
from machine import Pin
import time

led = Pin(0, Pin.OUT)
button = Pin(13, Pin.IN, Pin.PULL_UP)

state = False
prev = 1

while True:
    curr = button.value()

    if prev == 1 and curr == 0:
        state = not state
        led.value(state)
        time.sleep(0.3)

    prev = curr
```





5. Assume you are a light operator in a manually operated DJ setup. You have three racks of three LEDs each. Use a push button switch system to manually cycle through the panels by turning them on and off.

Code

```
from machine import Pin
import time

# 9 LEDs organized as 3 racks x 3 LEDs each
rack1 = [Pin(i, Pin.OUT) for i in [0, 1, 2]]   # Rack 1: GP0,1,2
rack2 = [Pin(i, Pin.OUT) for i in [3, 4, 5]]   # Rack 2: GP3,4,5
rack3 = [Pin(i, Pin.OUT) for i in [6, 7, 8]]   # Rack 3: GP6,7,8

btn = Pin(14, Pin.IN, Pin.PULL_UP)  # Single pushbutton

# State machine: 0-8 cycles through all panels
current_panel = 0
last_btn = 1
last_time = 0
debounce = 200  # DJ needs reliable button response

panels = [rack1, rack2, rack3]

print("DJ LIGHT PANELS ACTIVE")
print("Press button to cycle: Rack1→Rack2→Rack3→Rack1...")

while True:
    now = time.ticks_ms()

    # Button press detected (debounced for live performance)
    btn_val = btn.value()
    if btn_val == 0 and last_btn == 1 and time.ticks_diff(now, last_time) > debounce:

        # Turn OFF all previous panels
        for rack in panels:
```

```python
        for led in rack:
            led.value(0)

    # Activate NEW panel (all 3 LEDs in rack)
    current_panel = (current_panel + 1) % 3
    for led in panels[current_panel]:
        led.value(1)

    print(f" PANEL {current_panel+1} LIVE!")
    last_time = now

  last_btn = btn_val
  time.sleep_ms(10)
```

—--


```json
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 "editor": "wokwi",
 "parts": [
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    [ "r0:2", "l00:A", "green", [ "v0" ] ],
    [ "l00:C", "pico:GND.1", "black", [ "h0" ] ],
    [ "pico:GP1", "r1:1", "green", [ "v0" ] ],
    [ "r1:2", "l01:A", "green", [ "v0" ] ],
    [ "l01:C", "pico:GND.1", "black", [ "h0" ] ],
    [ "pico:GP2", "r2:1", "green", [ "v0" ] ],
    [ "r2:2", "l02:A", "green", [ "v0" ] ],
```

```
    [ "l02:C", "pico:GND.1", "black", [ "h0" ] ],
    [ "pico:GP3", "r3:1", "green", [ "v0" ] ],
    [ "r3:2", "l10:A", "green", [ "v0" ] ],
    [ "l10:C", "pico:GND.1", "black", [ "h0" ] ],
    [ "pico:GP4", "r4:1", "green", [ "v0" ] ],
    [ "r4:2", "l11:A", "green", [ "v0" ] ],
    [ "l11:C", "pico:GND.1", "black", [ "h0" ] ],
    [ "pico:GP5", "r5:1", "green", [ "v0" ] ],
    [ "r5:2", "l12:A", "green", [ "v0" ] ],
    [ "l12:C", "pico:GND.1", "black", [ "h0" ] ],
    [ "pico:GP6", "r6:1", "green", [ "v0" ] ],
    [ "r6:2", "l20:A", "green", [ "v0" ] ],
    [ "l20:C", "pico:GND.1", "black", [ "h0" ] ],
    [ "pico:GP7", "r7:1", "green", [ "v0" ] ],
    [ "r7:2", "l21:A", "green", [ "v0" ] ],
    [ "l21:C", "pico:GND.1", "black", [ "h0" ] ],
    [ "pico:GP8", "r8:1", "green", [ "v0" ] ],
    [ "r8:2", "l22:A", "green", [ "v0" ] ],
    [ "l22:C", "pico:GND.1", "black", [ "h0" ] ],
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    [ "btn:2.r", "pico:GND.1", "black", [ "h0" ] ]
  ],
  "dependencies": {}
}
```
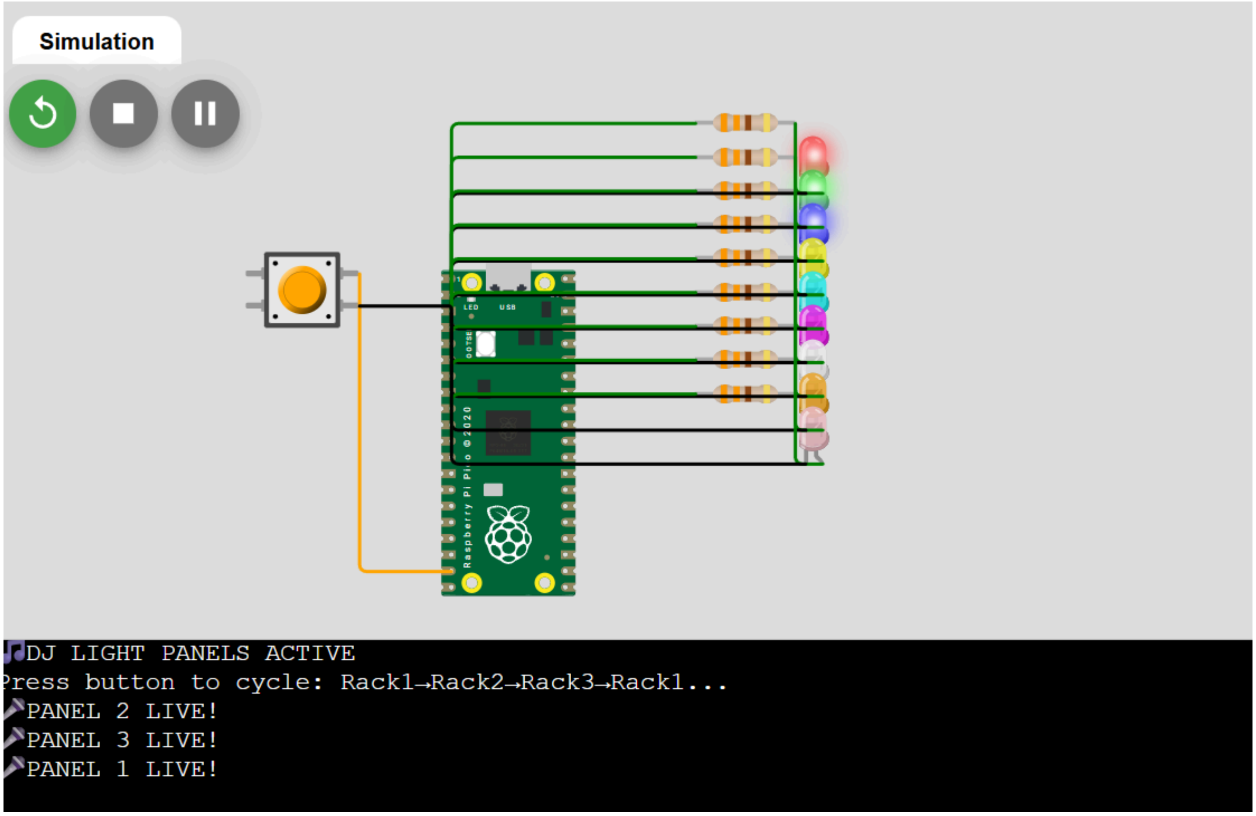
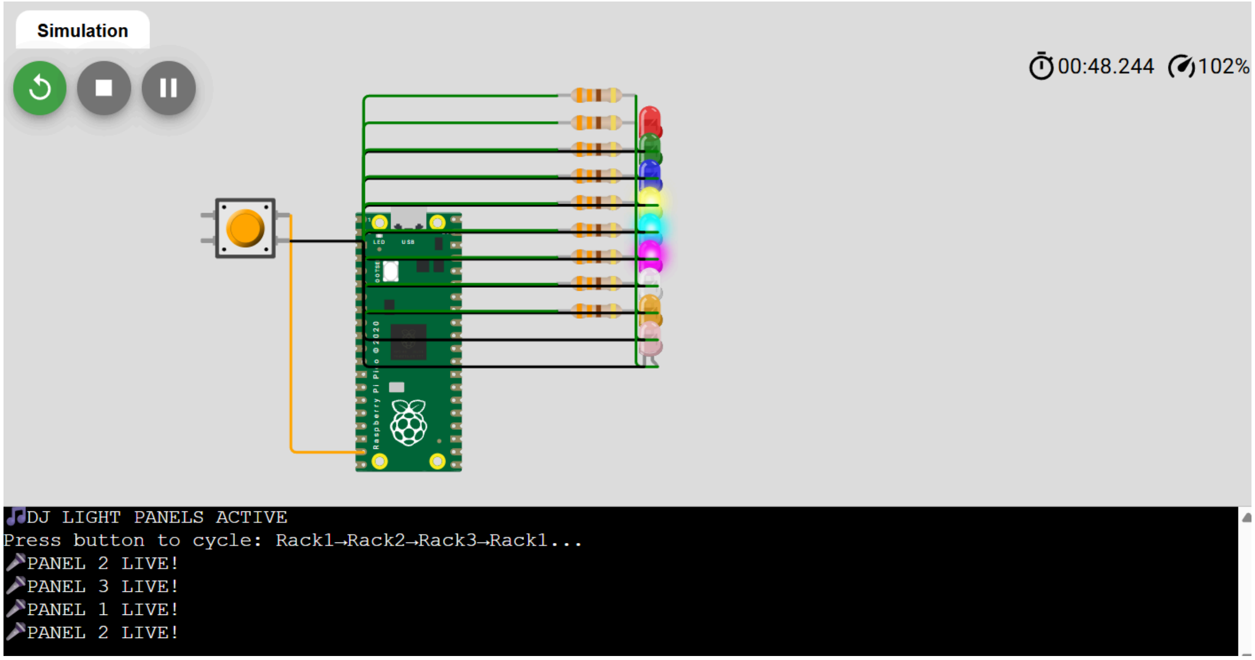
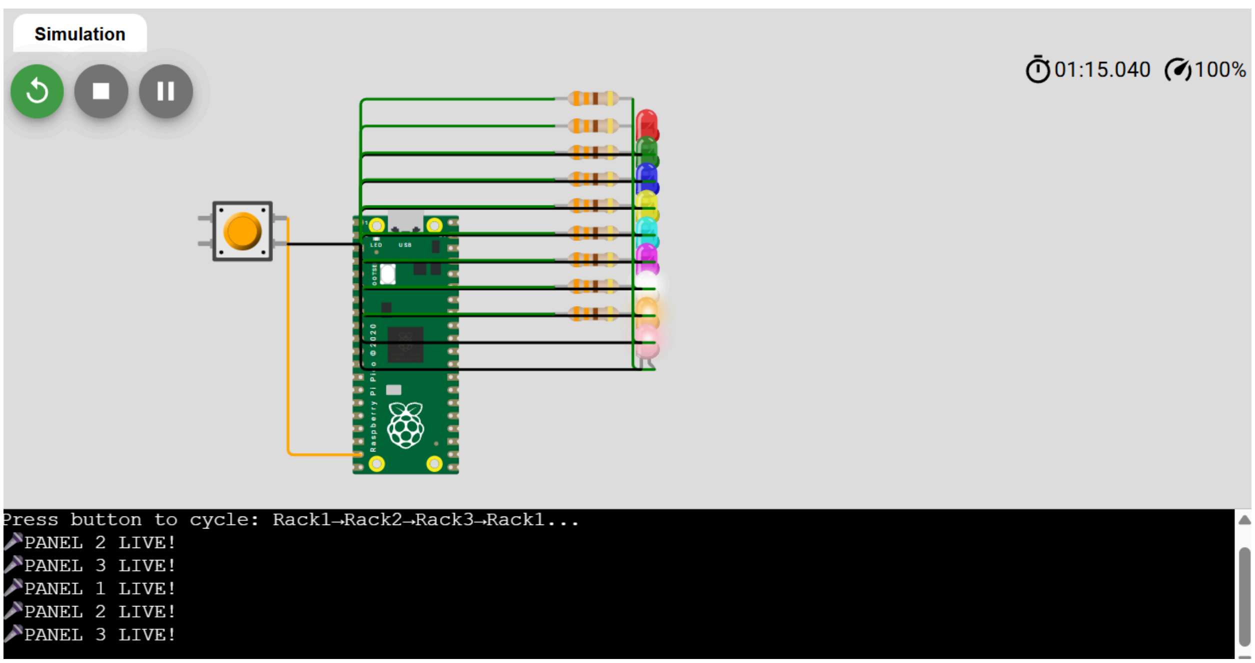


Simulation
🎵DJ LIGHT PANELS ACTIVE
Press button to cycle: Rack1→Rack2→Rack3→Rack1...
🎤PANEL 2 LIVE!
🎤PANEL 3 LIVE!
🎤PANEL 1 LIVE!





6. Design a state memory system which is able to remember the state of a button using variables. Use a combination of two switches to act as a decoder to switch on one of 4 LEDs

Code

```
from machine import Pin
import time
```

```python
s1 = Pin(14, Pin.IN, Pin.PULL_UP)  # Slide switch 1
s0 = Pin(15, Pin.IN, Pin.PULL_UP)  # Slide switch 2
clk = Pin(4,  Pin.IN, Pin.PULL_UP) # Orange button
leds = [Pin(i, Pin.OUT) for i in [0,1,2,3]]

state = 0
last_clk = 1
last_time = 0

print("SLIDE switches → PRESS ORANGE → Watch LED")

while True:
    now = time.ticks_ms()
    clk_val = clk.value()
    if clk_val == 0 and last_clk == 1 and time.ticks_diff(now,
last_time) > 50:
        state = ((1 - s1.value()) << 1) | (1 - s0.value())
        print(f"LATCHED: {state:02b} → LED{state}")
        last_time = now
    last_clk = clk_val
    for i in range(4): leds[i].value(1 if i == state else 0)
    time.sleep_ms(10)
```


{
 "version": 1,
 "editor": "wokwi",
 "parts": [
  {"type": "wokwi-pi-pico", "id": "pico", "top": 0, "left": 0},
  {"type": "wokwi-led", "id": "led0", "top": -60, "left": 250, "attrs": {"color": "red"}},
  {"type": "wokwi-led", "id": "led1", "top": -30, "left": 250, "attrs": {"color": "green"}},
  {"type": "wokwi-led", "id": "led2", "top": 0, "left": 250, "attrs": {"color": "blue"}},
  {"type": "wokwi-led", "id": "led3", "top": 30, "left": 250, "attrs": {"color": "yellow"}},
  {"type": "wokwi-resistor", "id": "r0", "top": -65, "left": 200, "attrs": {"value": "330"}},
  {"type": "wokwi-resistor", "id": "r1", "top": -35, "left": 200, "attrs": {"value": "330"}},
  {"type": "wokwi-resistor", "id": "r2", "top": -5, "left": 200, "attrs": {"value": "330"}},
  {"type": "wokwi-resistor", "id": "r3", "top": 25, "left": 200, "attrs": {"value": "330"}},

```
    {"type": "wokwi-slide-switch", "id": "sw1", "top": -50, "left": -150},
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"orange"}}
  ],
  "connections": [
    ["pico:GP0", "r0:1", "green", ["v0"]], ["r0:2", "led0:A", "green", ["v0"]], ["led0:C",
"pico:GND.1", "black", ["h0"]],
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"pico:GND.1", "black", ["h0"]],
    ["pico:GP2", "r2:1", "green", ["v0"]], ["r2:2", "led2:A", "green", ["v0"]], ["led2:C",
"pico:GND.1", "black", ["h0"]],
    ["pico:GP3", "r3:1", "green", ["v0"]], ["r3:2", "led3:A", "green", ["v0"]], ["led3:C",
"pico:GND.1", "black", ["h0"]],

    ["sw1:2", "pico:GP14", "blue", ["v0"]], ["sw1:3", "pico:GND.1", "black", ["h0"]],
    ["sw0:2", "pico:GP15", "purple", ["v0"]], ["sw0:3", "pico:GND.1", "black", ["h0"]],
    ["clk:1.r", "pico:GP4", "orange", ["v0"]], ["clk:2.r", "pico:GND.1", "black", ["h0"]]
  ],
  "dependencies": {}
}
```
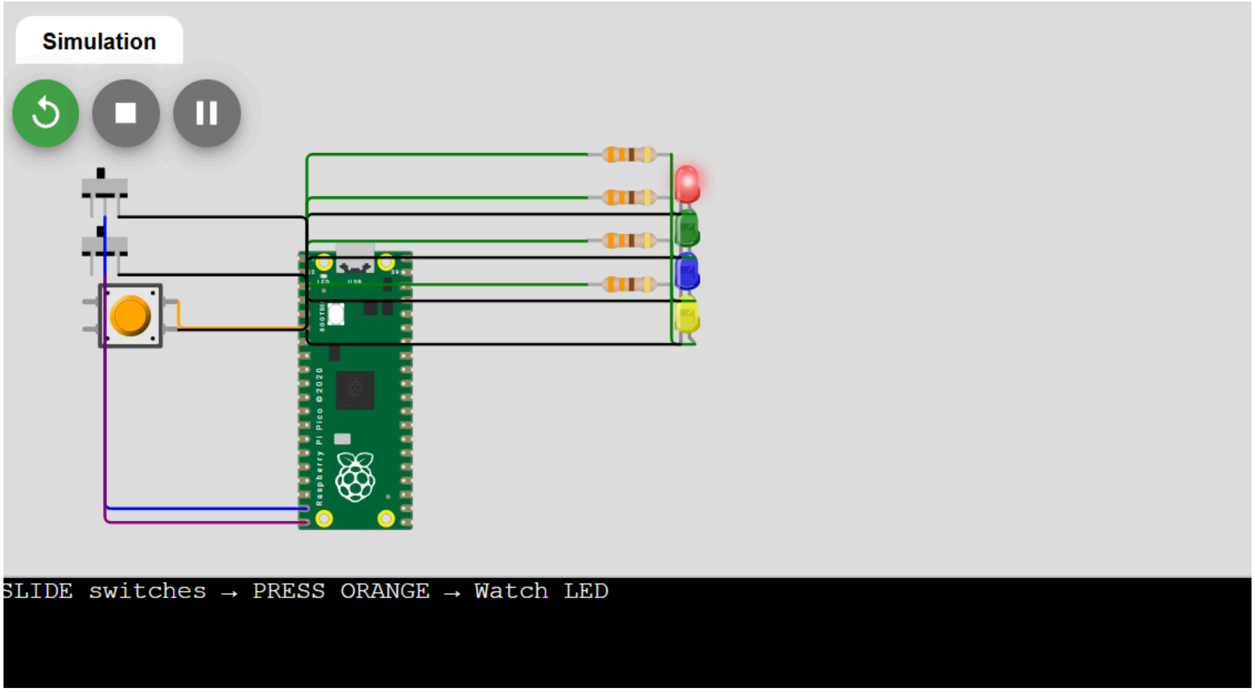
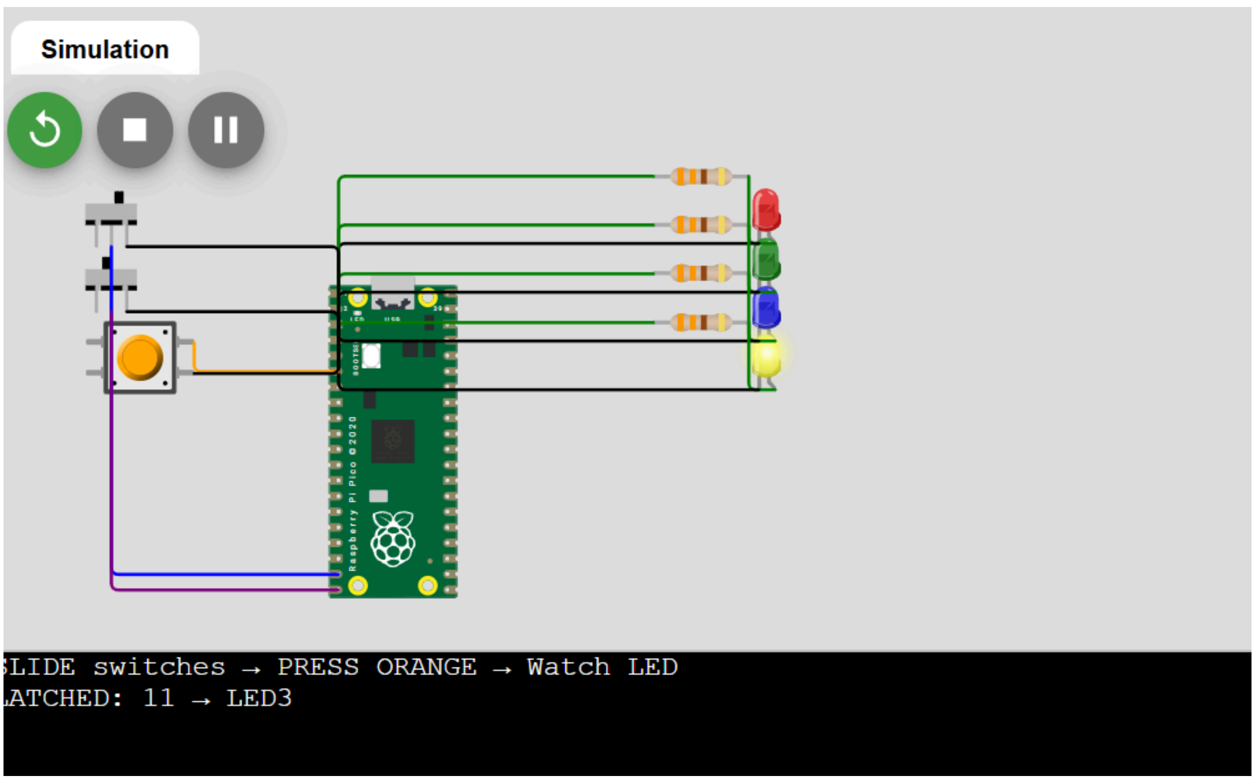
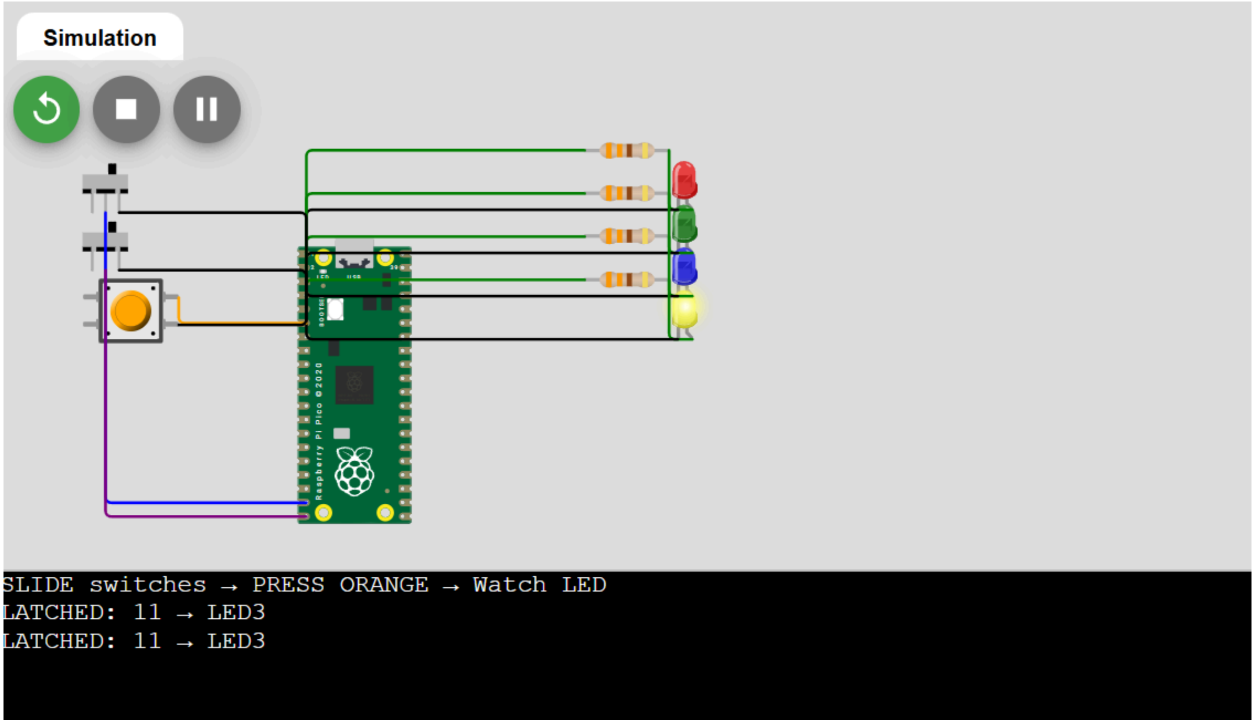
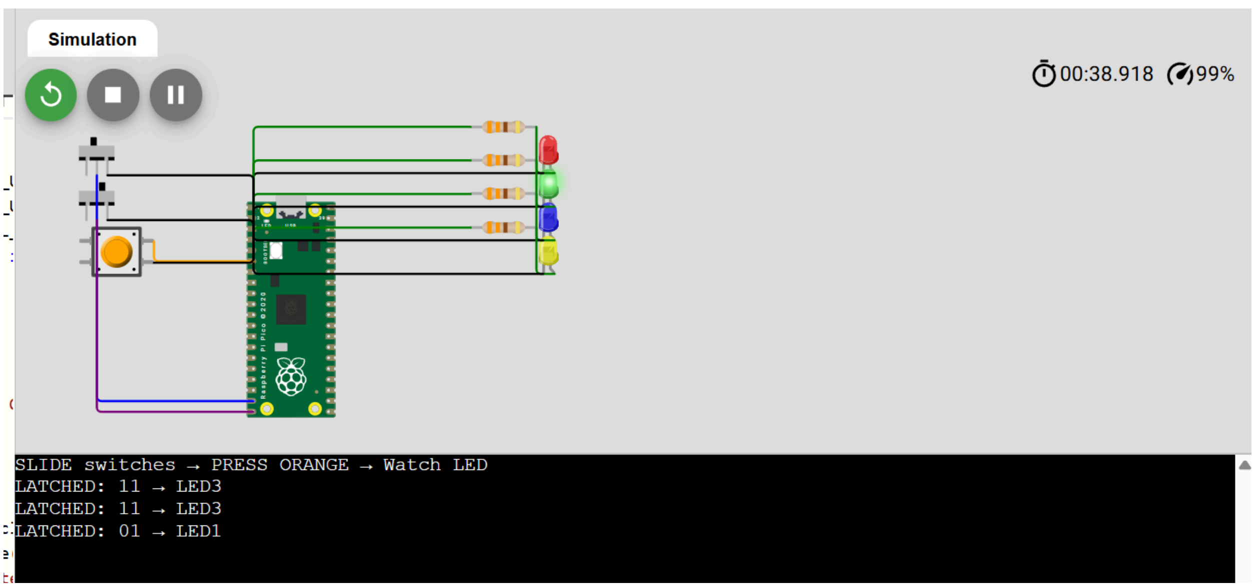


Simulation
SLIDE switches → PRESS ORANGE → Watch LED


Simulation
SLIDE switches → PRESS ORANGE → Watch LED
LATCHED: 11 → LED3


Simulation
SLIDE switches → PRESS ORANGE → Watch LED
LATCHED: 11 → LED3
LATCHED: 11 → LED3


Simulation
00:38.918
99%
SLIDE switches → PRESS ORANGE → Watch LED
LATCHED: 11 → LED3
LATCHED: 11 → LED3
LATCHED: 01 → LED1